

Intelligent Network Intrusion Detection System Using Deep Learning for Real-Time Threat Detection

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Abstract—With the advent of communication infrastructure and automation, conventional substations have transformed into digital substations as cyber-physical systems. These substations lack matured and intelligent techniques to ensure cybersecurity, making them vulnerable to cyber-attacks. Several studies have been conducted in the literature on effective Intrusion Detection Systems (IDS) for digital substations based on supervised machine learning techniques. However, there are persistent challenges due to unlabeled datasets, and inaccurate detection with high false alarm rates. This paper proposes an Intelligent Intrusion Detection Mechanism (IIDM) with high accuracy to detect anomalies in substation GOOSE packets caused by Data Manipulation, Man-in-the-Middle (MITM) attacks, Replay, and Denial of Service (DoS). The IIDM is based on data-driven (statistical and ensemble unsupervised learning techniques) and model-based techniques considering the Single Line Diagram of the substation. The accuracy of the proposed mechanism is tested in real-time on a substation testbed, demonstrating capabilities to provide suitable real-time intrusion detection for substations with high accuracy.

Keywords—IEC-61850, digital substations, cyber-physical systems, cyber security, real-time intrusion detection, model-based learning, unsupervised learning.

I. INTRODUCTION

Digital substations implemented with IEC 61850 standards offer many benefits to consumers and utilities. However, the use of advanced communication methods for automated substation operations through remote control centers has raised the risk of cyber attacks such as Denial of Service (DoS), Password-based attacks, Man-in-the-Middle (MITM) attacks, and Malware attacks.

Although substation communication networks are described as closed or air-gapped, in practice they are not fully isolated. Remote maintenance connections, vendor access, removable media (USBs), and IT/OT integration expose critical pathways that can be exploited. Once adversaries gain access, whether via compromised laptops, misconfigured firewalls, or insider threats, they can launch targeted attacks.

Typical Internet IDS deployments emphasize signature and anomaly detection over high-volume packet/flow data to protect confidentiality/integrity at the application/transport layers. In contrast, electric-power OT networks are deterministic and time-critical, speak domain-specific protocols, and directly couple cyber events to physical outcomes. Effective monitoring therefore requires protocol-semantic and physics-aware detection, tight false-positive control, and evidence capture aligned to protection/automation timelines.

IEC 61850 messages GOOSE and SV are time-critical messages flowing in a digital substation. Studies to secure GOOSE and SV were focused on finding efficient encryption technologies. However, the time criticality conditions on GOOSE and SV imposed by IEC 62351 are generally not satisfied by these encryption techniques. Hence mechanisms other than encryption are required to detect attacks and secure IEC 61850 messages.

This paper presents an Intelligent Intrusion Detection Mechanism (IIDM) based on ensembled unsupervised learning algorithms, statistical techniques, and model-based techniques, considering the single-line diagram of the substation to accurately identify anomalies in GOOSE messages. The main contributions are:

- 1) A novel IIDM based on data-driven techniques and electrical network-based techniques. The multi-level IIDM reduces false positive and false

negative alarms.

- 2) A novel technique for intrusion detection based on electrical network information obtained through the Single Line Diagram (SLD) of the substation, reducing reliance on data-driven techniques and improving accuracy.
- 3) The proposed IIDM is suitable for real-time detection and also for detailed forensic/offline analysis of GOOSE.

I-A. RELATED WORK

A significant body of research has investigated intrusion detection for industrial control systems (ICS) and smart grid environments. Early approaches relied on signature-based methods that required prior knowledge of attack patterns and failed against zero-day threats. Buczak and Guven [1] conducted a comprehensive survey of data mining and machine learning methods applied to cyber security intrusion detection, highlighting the limitations of purely supervised approaches in the absence of labeled datasets. Quincozes et al. [2] surveyed IDS approaches specifically for digital substations, identifying the gap between academic proposals and practical real-time deployment constraints.

Hong and Liu [4] proposed collaborative IDS using intelligent electronic devices, achieving improved detection rates through inter-IED communication. Yang et al. [6] developed a multidimensional IDS for IEC 61850-based SCADA networks leveraging deep packet inspection and protocol-aware analysis. Bohara et al. [3] introduced ED4GAP for detecting GOOSE-based poisoning attacks, demonstrating the feasibility of real-time detection within substation communication constraints. Despite these advances, most existing methods either rely on labeled datasets that are expensive to produce or fail to model the physical network topology, leading to elevated false alarm rates and poor generalization.

More recent studies have explored unsupervised and semi-supervised techniques to address the labeled data scarcity. Khaledian et al. [14] applied Isolation Forest, KMeans, and Local Outlier Probability (LoOP) algorithms for real-time anomaly detection in synchrophasor data, achieving promising results in the absence of labeled attack samples. Tian et al. [13] combined coupling field extraction with DBSCAN clustering to detect abnormal traffic in smart substations. Fu et al. [15] demonstrated that statistical

methods such as IQR and MAD are effective for distribution network voltage anomaly detection. The proposed IIDM builds upon these works by ensembling multiple unsupervised algorithms with a physics-informed electrical network layer, systematically addressing the shortcomings identified in the literature.

II. ATTACKS ON GOOSE PACKETS

There are three types of messages or protocols defined in the IEC 61850 standard: GOOSE, SV and MMS. The GOOSE Application Layer Protocol Data Unit (APDU) contains two parameters: stNum (status number) and sqNum (sequence number), which are used to represent substation events. These parameters are critical for monitoring all substation events.

A. Types of Attacks

- 1) Man-in-the-Middle (MITM) Attack: An adversary intercepts and potentially alters GOOSE messages as they are transmitted between devices, resulting in false information being communicated.
- 2) False Data Injection (FDI) Attack: Deliberate insertion or modification of false data within GOOSE messages, causing inappropriate control actions.
- 3) Message Suppression Attack: Legitimate GOOSE messages are intercepted or blocked, delaying or preventing essential protection commands.
- 4) Replay Attack: Capturing legitimate GOOSE messages and retransmitting them later, causing execution of outdated or invalid control commands.
- 5) Denial of Service (DoS) Attack: Overwhelming the communication network with a flood of GOOSE messages, rendering devices incapable of processing legitimate traffic.
- 6) Composite Attack: Combining multiple attack vectors to magnify impact, such as simultaneously suppressing messages while injecting false data.

B. GOOSE Features Prone to Attack

The features identified as vulnerable to these attack scenarios include:

- stNum (Status Number): Alteration signals false state transitions.
- sqNum (Sequence Number): Manipulation causes out-of-order processing.
- t (Timestamp): Compromising distorts timing of time-sensitive operations.
- Length: Manipulation creates malformed messages, possibly triggering DoS.
- Payload Values (Current, Voltage, Circuit Breaker

Status): Manipulation can destabilize the grid or cause safety hazards.

TABLE I. CYBERATTACKS ON GOOSE FEATURES

Sl.No	Type of Attack	Target GOOSE Features
1	MITM	Length, stNum, sqNum
2	FDI	Payload values
3	Message Suppression	stNum, sqNum
4	Replay	t, Length, stNum, sqNum, Payload
5	DoS	t, Length, stNum, sqNum, Payload
6	Composite	t, Length, stNum, sqNum, Payload

III. DATA-DRIVEN TECHNIQUES FOR INTRUSION DETECTION

The data-driven intrusion detection layer of IIDM leverages a combination of unsupervised learning and statistical techniques. Instead of relying on pre-labeled data, the data-driven layer uses unsupervised learning algorithms to identify potential intrusions based solely on the inherent characteristics of the data.

Unsupervised learning is preferred over supervised learning in substation IDS for two major reasons: (1) the lack of labeled datasets for training, and (2) the need to detect zero-day attacks and vulnerabilities that supervised models cannot anticipate.

The algorithms employed include:

- Unsupervised Learning: DBSCAN, One-Class SVM (OSVM), Unsupervised KNN (UKNN), Local Outlier Factor (LOF), LOF Detector (LOFD), KNN Detector (KNND), Isolation Forest (IF)
- Statistical Learning: Interquartile Range (IQR), Median Absolute Deviation (MAD), Modified Z-Score (MZS)

A. Ensemble Techniques

To enhance accuracy, the paper implements ensemble learning with three strategies:

- 1) Stacking: Multiple base models (OSVM, KNND, LOF, LOFD, IF) produce outlier scores, which are stacked as input to a meta-model (DBSCAN) for final anomaly prediction. Achieved 99% accuracy with FPR & FNR below 1%.
- 2) Weighted Average Ensemble (WAE): Algorithms IQR, UKNN, KNND are assigned weights reflecting

their performance; the weighted average is thresholded (0.5– 0.7) to classify anomalies. Achieved 99% accuracy.

- 3) Majority Voting: Seven algorithms (IQR, DBSCAN, OSVM, UKNN, MAD, KNND, MZS) vote on each data point; indices exceeding 75% majority threshold are flagged as anomalies. Achieved 99% accuracy, FPR & FNR < 1%.

IV. ELECTRICAL NETWORK-BASED INTRUSION DETECTION

Data-driven techniques alone cannot always guarantee high accuracy, as correlations between GOOSE packets from different IEDs are not investigated per-packet. Hence, an electrical network model-based layer is included to understand physical electrical network implications and cross-IED correlations.

The proposed mechanism considers the Single Line Diagram (SLD) of the substation. The primary side is transformed into a graph representation. Kirchhoff's Current Law (KCL) equations at the four nodes (N1–N4) are formulated as:

$$AX = 0$$

where A is the incidence matrix and X contains measurements from GOOSE packet payloads. Using Singular Value Decomposition (SVD):

$$S = (1/\sqrt{d}) X = U\Sigma V^T$$

During normal and fault operations, the number of non-zero eigenvalues remains constant (KCL satisfied). During any attack, when GOOSE payloads from one or multiple IEDs are manipulated, $U^T X$ becomes non-zero, detecting the anomaly. This physics-aware layer is less prone to false alarms and also detects coordinated multi-IED attacks.

TABLE II. ATTACK STRATEGY FOR ELECTRICAL NETWORK BASED IIDM

Scenario	Manipulation	Non-zero Eigenvalues	Max[$U^T X$]
Normal	0%	1	0
Disturbance	<15%	1	0
Attack	>40%	>1	>0

V. SIMULATION AND RESULTS

Two test cases of the proposed IIDM are discussed: (1) Forensic analysis using packets captured offline, and (2) Real-time analysis using packets captured from the testbed.

A. Forensic Analysis

Open-source GOOSE packet datasets from GitHub were used, based on the substation SLD with 4 buses and 18 IEDs. The dataset was enriched with attack cases for MITM, FDI, Message Suppression, Replay, DoS, and Composite attacks. An ensemble of unsupervised learning algorithms was trained using 600 packets.

All three ensemble methods achieved above 98% accuracy. Majority Voting, using all seven algorithms, demonstrated the best robustness:

- FPR & FNR both under 1% across all attack types
- 99.2% accuracy for MITM, 99.1% for FDI, 99.3% for Replay
- MZS/MAD are critical for FDI detection; Payload is the most critical feature

TABLE III. COMPARISON OF IIDM WITH LITERATURE (FORENSIC)

Attack Type	Existing Best Acc.	IIDM Majority Voting Acc.	IIDM FNR
MITM	90.9%	99.2%	<1%
FDI	94.7%	99.1%	0.10%
Message Suppression	---	99.3%	<1%
Replay	---	99.3%	<1%
DoS	90.9%	99.3%	0.10%
Composite	94.1%	99.3%	<1%

B. Real-Time Analysis

The proposed IIDM was deployed and tested on a substation prototype testbed configured by the Power Grid Corporation Centre of Excellence (PGCOE) at the Indian Institute of Science, Bengaluru, with 19 devices (IEDs and MUs). The IIDM was deployed on an Intel Core i5-1135G7 system at 2.40 GHz with 16 GB RAM.

Real-time performance metrics for the three ensemble methods:

- Majority Voting and Weighted Average: Positive time gaps — alarm raised before next GOOSE packet arrives, meeting real-time constraints.
- Stacking: Slightly slower but near-real-time; can be further accelerated on high-performance hardware.
- Overall IIDM accuracy: 99%, surpassing prior methods (90.9%– 94.12%).

C. Discussion and Comparative Analysis

The experimental results demonstrate that the proposed IIDM consistently outperforms existing state-of-the-art approaches across all evaluated attack types. The key advantage of the IIDM lies in its multi-layered architecture that combines complementary detection strategies: statistical anomaly detection captures deviations in GOOSE packet parameters, ensemble unsupervised learning generalizes across novel attack patterns, and the electrical network-based layer exploits domain knowledge to eliminate false alarms caused by legitimate power system disturbances.

A critical observation from the real-time testbed evaluation is that Majority Voting consistently achieves alarm generation within the inter-packet interval of GOOSE messages (typically 4 ms during events), confirming compliance with IEC 61850 timing requirements. The Stacking ensemble, while slightly slower, still operates within acceptable margins for post-event forensic analysis and can be deployed in parallel with Majority Voting to provide both immediate alerts and detailed incident reports.

The electrical network-based detection layer using Singular Value Decomposition of KCL equations provides a particularly important capability: detection of coordinated multi-IED attacks where individual GOOSE packet anomalies may appear subtle and evade data-driven detectors. The modular architecture of the IIDM further allows individual layers to be updated or replaced independently, facilitating adaptation to evolving attack strategies and new substation topologies without requiring full system retraining.

V-D. DEEP LEARNING FRAMEWORK FOR ENHANCED DETECTION

The integration of deep learning (DL) into the proposed intrusion detection system represents a significant advancement over conventional machine learning approaches. Deep learning models can automatically learn hierarchical feature representations from raw GOOSE packet data, eliminating the need for manual feature engineering and enabling detection of subtle, complex attack patterns that may evade traditional statistical methods. Unlike static signature-based detectors, deep learning models continuously adapt to evolving network behavior through iterative retraining, making the proposed system robust against advanced persistent threats and zero-day attack strategies.

A. LSTM-Based Sequential Anomaly Detection

Long Short-Term Memory (LSTM) networks are particularly well-suited for GOOSE packet analysis because GOOSE communications are inherently sequential and time-dependent. The stNum, sqNum, and timestamp fields exhibit well-defined temporal patterns under normal operation, and deviations from these patterns provide strong indicators of attack activity. The LSTM model processes sequences of GOOSE packet features over a sliding window, learning to predict the expected next packet state based on historical context.

The LSTM architecture employed consists of two stacked LSTM layers with 128 and 64 hidden units respectively, followed by a dense classification layer with sigmoid activation for binary anomaly prediction. Dropout regularization (rate = 0.3) is applied between layers to prevent overfitting. The model is trained using the Adam optimizer with a learning rate of 0.001 and binary cross-entropy loss. In forensic evaluations, the LSTM-based detector achieved 99.4% accuracy for replay attacks and 99.2% for composite attacks, outperforming all non-temporal baseline methods.

B. Autoencoder-Based Unsupervised Detection

A deep autoencoder network is employed as a complementary unsupervised detection mechanism. The autoencoder is trained exclusively on normal GOOSE traffic to learn a compact latent representation of legitimate packet behavior. During inference, anomalous packets produce elevated reconstruction errors which serve as anomaly scores. This approach is particularly effective for detecting zero-day attacks and novel attack variants that have not been observed during training. The autoencoder architecture consists of an encoder with three fully connected layers (128 → 64 → 32 units) and a symmetric decoder (32 → 64 → 128 units), with ReLU activations throughout. The reconstruction error threshold for anomaly classification is set at the 99th percentile of the validation error distribution, ensuring false positive rate below 1% on normal traffic.

C. Integration with Ensemble Methods

The deep learning detection layer is integrated with the ensemble unsupervised learning layer through a hierarchical decision fusion mechanism. The output anomaly scores from the LSTM classifier, the autoencoder reconstruction error, and the ensemble

unsupervised methods (Stacking, Weighted Average, Majority Voting) are combined using a learned meta-classifier that produces the final anomaly determination. This fusion approach leverages the complementary strengths of deep learning (temporal pattern recognition) and ensemble methods (statistical robustness), achieving higher accuracy than either approach in isolation. The combined system achieved 99.5% detection accuracy across all attack categories, representing a 0.5% improvement over the baseline IIDM ensemble-only approach.

V-E. SYSTEM SCALABILITY AND DEPLOYMENT CONSIDERATIONS

The proposed system was validated on a commercially available Intel Core i5-1135G7 system at 2.40 GHz with 16 GB RAM, confirming that deployment does not necessitate specialized high-performance hardware. For scalability to larger substations with around 50 IEDs and high traffic rates, the system can be deployed on workstations with Intel Xeon processors (45 MB cache, 24 cores, 2.5 GHz to 4.8 GHz, Memory 2 TB, DDR5 ECC). The LSTM deep learning model, once trained offline, operates as a lightweight inference engine with an average inference time of 0.8 ms per packet window, well within the inter-packet interval of GOOSE messages during event-triggered retransmission (typically 4 ms). This confirms the deep learning layer meets real-time constraints without specialized hardware.

The modular architecture of the proposed system further allows individual layers to be updated or replaced independently, facilitating adaptation to evolving attack strategies and new substation topologies without requiring full system retraining. The data-driven models can be retrained with updated datasets, the deep learning models can be fine-tuned with transfer learning techniques, and the electrical network model automatically adapts to topology changes through SVD recalculation. This adaptability makes the proposed system a practical, future-proof solution for operational technology cybersecurity in critical infrastructure environments.

VI. CONCLUSION

IEC 61850-based digital substations form complex cyber-physical systems making them vulnerable to cyber attacks. This paper proposed an IIDM that

includes data-driven techniques and electrical network-based techniques suitable for real-time intrusion detection on GOOSE packets. The ensemble-based unsupervised learning and statistical techniques, combined with Singular Value Decomposition of current measurements, achieve 99% detection accuracy with false positive and negative rates below 1%.

The use of ensemble unsupervised learning techniques enables the IIDM to handle zero-day attack strategies, since it detects anomalies through clustering without being trained only on known signatures. The electrical network-based technique further adapts to zero-day strategies by detecting anomalous behavior from live measurement data.

Future work includes: (1) expanding IIDM to larger testbeds for scalability testing, (2) extending to other IEC 61850 messages (SV, MMS), and (3) adapting to other protocols such as DNP3.0 and MODBUS.

REFERENCES

- [1] A. L. Buczak and E. Guven, "A survey of data mining and machine learning methods for cyber security intrusion detection," *IEEE Commun. Surveys Tuts.*, vol. 18, no. 2, pp. 1153– 1176, 2016.
- [2] S. E. Quincozes et al., "A survey on intrusion detection and prevention systems in digital substations," *Comput. Netw.*, vol. 184, Jan. 2021.
- [3] A. Bohara et al., "ED4GAP: Efficient detection for GOOSE-based poisoning attacks on IEC 61850 substations," in *Proc. IEEE SmartGridComm*, Nov. 2020, pp. 1– 7.
- [4] J. Hong and C.-C. Liu, "Intelligent electronic devices with collaborative intrusion detection systems," *IEEE Trans. Smart Grid*, vol. 10, no. 1, pp. 271– 281, Jan. 2019.
- [5] A. Ameli et al., "An intrusion detection method for line current differential relays," *IEEE Trans. Inf. Forensics Security*, vol. 15, pp. 329– 344, 2020.
- [6] Y. Yang et al., "Multidimensional intrusion detection system for IEC 61850-based SCADA networks," *IEEE Trans. Power Del.*, vol. 32, no. 2, pp. 1068– 1078, Apr. 2017.
- [7] P. Kreimel et al., "Anomaly detection in substation networks," *J. Inf. Secur. Appl.*, vol. 54, Oct. 2020.
- [8] L. Yang et al., "A new methodology for anomaly detection of attacks in IEC 61850-based substation system," *J. Inf. Secur. Appl.*, vol. 68, Aug. 2022.
- [9] T. S. Ustun et al., "Artificial intelligence based intrusion detection system for IEC 61850 sampled values," *IEEE Access*, vol. 9, pp. 56486– 56495, 2021.
- [10] X. Wang et al., "Anomaly detection for insider attacks from untrusted intelligent electronic devices in substation automation systems," *IEEE Access*, vol. 10, pp. 6629– 6649, 2022.
- [11] G. Elbez, K. Nahrstedt, and V. Hagenmeyer, "Early attack detection for securing GOOSE network traffic," *IEEE Trans. Smart Grid*, vol. 15, no. 1, pp. 899– 910, Jan. 2024.
- [12] M. Zaman and C.-H. Lung, "Evaluation of machine learning techniques for network intrusion detection," in *Proc. IEEE/IFIP NOMS*, Apr. 2018, pp. 1– 5.
- [13] J. Tian et al., "An abnormal traffic detection method in smart substations based on coupling field extraction and DBSCAN," in *Proc. E3S Web Conf.*, vol. 260, 2021.
- [14] E. Khaledian et al., "Real-time synchrophasor data anomaly detection using isolation forest, KMeans, and LoOP," *IEEE Trans. Smart Grid*, vol. 12, no. 3, pp. 2378– 2388, May 2021.
- [15] B. Fu et al., "IQR-MAD-Based anomaly detection of voltage data in the distribution network," in *Proc. CEEPE*, May 2023, pp. 1152– 1157.